MILITARY SPECIFICATION

DETECTOR/COOLER ASSEMBLY FOR AN/PAS-7

This specification is approved for use by USA Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detector/cooler assembly for the AN/PAS-7 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallet, Materials Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	- Strapping, Steel, Flat and Seals
PPP-B-585	- Boxes, Wood, Wirebound
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock Corner
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple Wall
PPP-C-843	- Cushioning Material, Cellulosic
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	- Strapping, Nonmetallic, (and Connectors)
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced

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MIL-P-116	- Preservation-Packaging Methods of
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-B-131	- Barrier Materials, Water Vaporproof, Flexible, Heat-Sealable
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	- Marking of Electronic Items

STANDARDS

FEDERAL

FED-STD-356	- Commercial Packaging of Supplies and Equipment

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-171	- Finishing of Metal and Wood Surfaces
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SM-D-774663	- Detector/Cooler

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Item definition. The detector/cooler assembly is one of the components of the Viewer Infrared, AN/PAS-7. The detector/cooler assembly consists of a linear array of 48 photoconductive lead selenide (PbSe) elements in a flat plane, and a thermoelectric cooler, mounted in a dewar. The array shall operate in the 3.0 to 5.0 micron spectral band.

3.2 First article. When specified the contractor shall furnish sample unit(s) for first article inspection and approval, (see 4.3 and 6.2).

3.3 Construction. Unless otherwise specified, the detector/cooler, assembly shall be constructed in accordance with USAECOM Drawing SM-D-774663.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Material not specified, shall be selected by the contractor, shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

3.5 Operational and environmental.

3.5.1 Operating temperature. The thermoelectric cooler shall cool the detector array to its operating temperature. This temperature shall be no greater than 213° Kelvin (K).

3.5.2 Detectivity. With a load impedance of 1 megohm, ±5 percent, a bias voltage of less than or equal to 20 volts direct current (Vdc), 500° or 1000° K blackbody temperature, 295 background and an equivalent noise bandwidth of not less than 4000 hertz (Hz); the peak detectivities shall be as follows:

a. 43 elements greater than or equal to 1.5×10^{10} cm $(Hz)^{\frac{1}{2}}$ $watt^{-1}$
b. 46 elements greater than or equal to 1.3×10^{10} cm $(Hz)^{\frac{1}{2}}$ $watt^{-1}$
c. All elements greater than or equal to 1.0×10^{10} cm $(Hz)^{\frac{1}{2}}$ $watt^{-1}$

3.5.3 Resistance. When cooled to their operating temperature the dc resistance of each detector element shall not be less than 10 megohms nor greater than 100 megohms.

3.5.4 Peak responsivity. The peak responsivity shall be equal to or greater than 2×10^5 volts/watt with a load impedance of 1 megohm, ±5 percent.

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3.5.5 Time constant. The time constant, defined as the time required for detector element response to reach 63 percent, ±1 percent of its steady state value after being irradiated by a step input of radiation in the 3 to 5 micron spectral region, shall not exceed 50 microseconds.

3.5.6 Cross talk. The cross talk between any two detector elements shall not be greater than 5 percent of the signal level.

3.5.7 Vacuum integrity. The detector/cooler assembly shall be hermetically sealed and shall be capable of complying with all operational requirements specified herein for a period of at least one year.

3.5.8 Spectral sensitivity. The detector array shall be sensitive in the 3 to 5 micron spectral region. The peak voltage responsivity shall occur at 5.0, ±.5 microns with the array cooled to its operating temperature.

3.5.9 Power.

3.5.9.1 Input voltage. The thermoelectric cooler, as part of the detector/cooler assembly, shall operate over an input voltage range of +5.5 to +7.8 volts dc.

3.5.9.2 Power consumption. Power consumption of the thermoelectric cooler, as part of the detector/cooler assembly, shall be no greater than 7.2 watts, at an input voltage of +6 Vdc, +0, -0.1 volts.

3.5.10 Temperature sensor. The temperature sensing element attached to the lower side of the top stage of the detector/cooler shall permit measurement of the temperature of the detector/cooler array to an accuracy of ±2° K within the temperature range of 185° to 223° K.

3.5.11 Optical window. The optical window aperture shall accept a converging beam having a total cone angle of 70°, ±5°. The average transmission of the window over the spectral region of 3 to 5 microns shall not be less than 85 percent.

3.5.12 Storage life. The storage life of the detector/cooler assembly shall not be less than one year at temperatures of -62°C to +68°C and barometric pressures of 725 to 875 millimeters of mercury (Hg) (sea level to 50,000 feet above sea level). At any time within the one year period the detector/cooler shall be capable of complying with the requirements specified herein.

3.5.13 Weight. The weight of the detector/cooler assembly shall be 0.32 pound maximum.

3.5.14 Temperature.

3.5.14.1 Operating. Minus 32°C to plus 54°C.

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3.5.14.2 Non-operating. Minus 62°C to plus 68°C.

3.5.15 Temperature shock. Minus 57°C, plus 0°C, minus 1.5°C, to plus 68°C, plus 1.5°C, minus 0°C.

3.5.16 Vibration. The detector/cooler assembly shall withstand vibration displacement of 0.060 inch peak to peak minimum over a frequency range of 10 to 55 Hz.

3.6 Government furnished equipment. Unless otherwise specified in the contract or purchase order, there shall be no Government furnished equipment provided.

3.7 Treatment and painting. Unless otherwise specified, the metallic parts of the detector/cooler assembly shall be given a treatment and painting per MIL-STD-171.

3.8 Marking. Unless otherwise specified, the detector/cooler assembly shall be marked per MIL-M-13231.

3.9 Workmanship. Unless otherwise specified workmanship of the detector/cooler assembly shall conform to Requirement 9, MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
c. Inspection of preparation for delivery (see 4.7)

4.3 First article inspection.

4.3.1 Inprocess inspection prescribed by a specification or drawing. Each lot of components and subassemblies shall have all parameters examined for defects in conformance with the inspection plan in the applicable specification or on the drawing.

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4.3.2 Inprocess inspection not prescribed by a specification or drawing. Each lot of components and subassemblies not prescribed by a specification or a drawing with a specific inspection plan shall have all parameters examined for defects in accordance with the sample plan tables of MIL-STD-105 with major defects (see 4.3.2.1) having an AQL of one percent defective combined and minor defects having AQL of 4 percent defective combined. General inspection level II shall be applicable to both sample plans.

4.3.2.1 Classification of defects for inprocess inspection not prescribed by a specification or drawing.

MAJOR

101. Performance requirements specified in notes appearing on drawings.
102. Linear or diametral tolerances of 0.002 inch or less.
103. Angular dimensions having a tolerance of 15 minutes of arc or less.
104. Machine finishes of 32 micro-inches RMS or finer.
105. Tolerances of position or form 0.002 inch or less.
106. Screw thread Class 3 or better.
107. Load requirements for springs.
108. Lubrication requirements including dry lubricant finishes.

MINOR

Defects not listed as major shall be classified as minor.

4.3.3 Examination. Each assembled detector/cooler assembly shall be examined for defects in accordance with Table I. One or more defects shall be cause for rejection.

TABLE I. Examination.

Inspection	Requirement paragraph	Inspection criteria
Any part or component missing or damaged	3.3, 3.4	MIL-STD-252
Treatment and painting not as specified	3.7	MIL-STD-171
Markings not as specified or illegible	3.8	MIL-M-13231
Workmanship not as specified	3.9	MIL-STD-454

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4.3.4 Tests. Upon successful completion of the inprocess inspections and assembled examination specified in 4.3 the first article detector/cooler assembly shall be subjected to the tests in Table II. Tests shall be conducted in the order listed. Failure of any test shall constitute first article failure.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Operating temperature	3.5.1	4.6.1
Detectivity	3.5.2	4.6.2
Resistance	3.5.3	4.6.3
Peak responsivity	3.5.4	4.6.4
Time constant	3.5.5	4.6.5
Cross talk	3.5.6	4.6.6
Spectral sensitivity	3.5.8	4.6.8
Input voltage	3.5.9.1	4.6.9.1
Power consumption	3.5.9.2	4.6.9.2
Temperature sensor	3.5.10	4.6.10
Optical window	3.5.11	4.6.11
Weight	3.5.13	4.6.13
Temperature operating	3.5.14.1	4.6.14
Temperature non-operating	3.5.14.2	4.6.14
Temperature shock	3.5.15	4.6.15
Vibration	3.5.16	4.6.16

4.4 Quality conformance inspection.

4.4.1 Inprocess inspection. Inprocess inspection on the detector/cooler assembly shall be as specified in 4.3.1 and 4.3.2.

4.4.2 Examination. Each assembled detector/cooler assembly shall be examined as specified in Table I. Presence of one or more defects shall be cause for rejection of that detector/cooler assembly.

4.4.3 Tests.

4.4.3.1 Group A inspection. Following successful completion of the inprocess inspection and assembly examination of 4.4.1 and 4.4.2, each detector/cooler assembly shall be tested in accordance with Table III in the order listed. Failure of any test shall be cause for rejection of that unit.

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TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Operating temperature	3.5.1	4.6.1
Detectivity	3.5.2	4.6.2
Resistance	3.5.3	4.6.3
Peak responsivity	3.5.4	4.6.4
Spectral sensitivity	3.5.8	4.6.8

4.4.3.2 Group B inspection. Group B tests shall be conducted on detector/cooler assemblies selected from units which have passed the tests in 4.4.3.1. Sampling of lots shall be in accordance with MIL-STD-105, AQL 2.5 percent defective, inspection level S-3 combined. Tests shall be conducted in accordance with Table IV, and shall be performed in the order listed.

TABLE IV. Group B inspection.

Test	Requirement paragraph	Test paragraph
Time constant	3.5.5	4.6.5
Cross talk	3.5.6	4.6.6
Power consumption	3.5.9.2	4.6.9.2
Temperature sensor	3.5.10	4.6.10
Optical window	3.5.11	4.6.11

4.4.3.3 Group C inspection. Group C inspection shall be conducted on detector/cooler assemblies selected from units which have passed the tests in 4.4.3.1. The sample(s) shall be subjected to the inspections specified in Table V. Sample(s) shall be selected in accordance with 4.4.3.3.1. Tests shall be performed in the order listed.

4.4.3.3.1 Sampling for Group C inspection.

4.4.3.3.1.1 Subgroup 1. For this subgroup 10 percent (1 minimum to 5 maximum) detector/cooler assembly shall be selected from the first quality conformance inspection production lot. For subsequent Group C inspection 10 percent (1 minimum to 5 maximum) detector/cooler assembly shall be selected every month from the monthly specified quantity (see 6.2).

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4.4.3.3.1.2 Subgroup 2. For this subgroup 1 detector/cooler assembly shall be selected from the first quality conformance inspection production lot. For subsequent Group C inspection 1 detector/cooler assembly shall be selected every third month from the third monthly specified quantity (see 6.2).

4.4.3.3.1.3 Subgroup 3. For this subgroup 1 detector/cooler assembly shall be selected from every sixth quality conformance inspection lot (see 6.2).

TABLE V. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Weight	3.5.13	4.6.13
Temperature operating	3.5.14.1	4.6.14
Vibration	3.5.16	4.6.16
Subgroup 2		
Temperature non-operating	3.5.14.2	4.6.14
Temperature shock	3.5.15	4.6.15
Subgroup 3		
Input voltage	3.5.9.1	4.6.9.1

4.4.3.3.2 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or order (see 6.2).

4.4.3.4 Reinspection of Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed Groups B and C inspections may be accepted on contract or order provided all damage is repaired and the units are resubjected to and pass Group A inspection.

4.5 Test methods. Tests required to determine compliance with the requirements of this specification shall be conducted in accordance with the procedures specified herein. Unless otherwise specified all tests shall be run at +23°C. (See 6.3).

4.6 Operational and environmental.

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4.6.1 Operating temperature. Operate the detector/cooler assembly and measure the resistance of the temperature sensor. Compare resistance with the temperature versus resistance calibration curve for unit under test, (see 4.6.10). Failure to comply with the requirement of 3.5.1 shall constitute a failure of this test.

4.6.2 Detectivity. Each channel of the array shall be measured. The array shall be irradiated with radiation from a 500° K or 1000° K, ±2° K blackbody and the radiation shall be chopped at 500 Hz, ±25 Hz. The background temperature shall be 295° K, ±3° K. Noise measurements shall be made over an equivalent noise bandwidth of 4000 Hz with a low frequency bandpass cut on of not greater than 20 Hz. The measured blackbody detectivities and the spectral response data shall be used to calculate the peak detectivities. All calculations, formulas, and conversion factors shall be supplied with the test data. Failures to comply with the requirement of 3.5.2 shall constitute failure of this test.

4.6.3 Resistance. Measure and record the bias current and voltage for each detector element when the detector elements are cooled to their operating temperature. The resistance of each detector element shall be calculated. Failure to comply with the requirement of 3.5.3 shall constitute a failure of this test.

4.6.4 Peak responsivity. The data obtained during the performance of the detectivity test (see 4.6.2) shall be used to calculate peak responsivity. The formulas and conversion factors to be used in the calculations shall be as described in 6.3.1. Failure to comply with the requirement of 3.5.4 shall constitute a failure of this test.

4.6.5 Time constant. Using a blackbody source that provides a signal to noise ratio between 9:1 and 11:1 and a chopper, the signal for each element selected shall be measured as a function of chopper frequency. As frequency is increased, the frequency at which the signal amplitude drops to 0.707 times its low frequency value shall be recorded. From this value, the time constant shall be calculated as described in 6.3.2. This test shall be made on one element selected at random from each third of the array, (three elements in the array shall be tested). This test shall be performed with the detector in an ambient temperature of 298° K, ±3° K. Failure to comply with the requirement of 3.5.5 shall constitute a failure of this test.

4.6.6 Cross talk. An optically focused spot, nominally 0.0015 inch in diameter (see 6.3.3), from a blackbody source of at least 500° K and chopped at a frequency between 500 and 600 Hz shall be made to impinge on the detector array. A signal to noise ratio of at least 10 within a 50 Hz electrical bandwidth centered at the source chopping frequency shall be produced in the output of the irradiated detector.

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The signal from each non-irradiated element shall be measured. This test shall be made on one element selected at random from each third of the array (three elements in the array shall be tested). This test shall be performed with the detector in an ambient temperature of 298° K, ±3° K. Failure to comply with the requirement of 3.5.6 shall constitute a failure of this test.

4.6.7 Vacuum integrity. The detector/cooler assembly shall comply with the requirement of 3.5.7 (see 6.2).

4.6.8 Spectral sensitivity. The spectral response shall be measured on three representative elements of the array. Failure to comply with the requirement of 3.5.8 shall constitute failure of this test.

4.6.9 Power.

4.6.9.1 Input voltage. With the thermoelectric cooler operating over an input voltage range of +5.5 to +7.8 volts dc, the detector/cooler assembly shall comply with all operational requirements when subjected to operational tests (see 4.6.1 through 4.6.6 and 4.6.8 through 4.6.11). Failure to comply with the requirements of 3.5.9.1 shall constitute a failure of this test.

4.6.9.2 Power consumption. Input dc voltage and current to the thermoelectric cooler shall be measured. Power consumption shall be calculated. Failure to comply with the requirement of 3.5.9.2 shall constitute a failure of this test.

4.6.10 Temperature sensor. A temperature versus resistance calibration curve, identified by detector/cooler assembly serial number, shall be provided for each detector/cooler assembly. The manufacturer shall certify the accuracy of each calibration curve. Failure to comply with the requirement of 3.5.10 shall constitute a failure of this test.

4.6.11 Optical window. The detector/cooler assembly and all pertinent substantiating data and drawings shall be examined to verify compliance with the requirements. Failure to comply with the requirement of 3.5.11 shall constitute a failure of this test.

4.6.12 Storage life. The detector/cooler assembly shall comply with the requirement of 3.5.12 (see 6.2).

4.6.13 Weight. Place the detector/cooler on a balance and measure its weight. Failure to comply with the requirement of 3.5.13 shall constitute a failure of this test.

4.6.14 Temperature.

4.6.14.1 High temperature. The detector/cooler assembly shall be subjected to test in accordance with Method 501, Procedure 1 of MIL-STD-

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810. Temperature for Step 2 shall be +68°C. Temperature for Step 4 shall be +49°C. Failure to operate or pass inspection of MIL-STD-810 shall constitute a failure of this test.

4.6.14.2 Low temperature. The detector/cooler assembly shall be subjected to test in accordance with Method 502, Procedure I of MIL-STD-810. The temperature for Step 2 shall be -62°C and shall be maintained until temperature stabilization of the test item is reached. The temperature for Step 4 shall be -32°C. Failure to operate or pass inspection of MIL-STD-810 shall constitute a failure of this test.

4.6.15 Temperature shock. The detector/cooler assembly shall be subjected to test in accordance with Method 503, Procedure I of MIL-STD-810. The maximum temperature for Steps 1 and 4 shall be +68°C. Failure to operate or pass inspection of MIL-STD-810 shall constitute a failure of this test.

4.6.16 Vibration. The detector/cooler assembly shall be operated under standard ambient conditions and a record made of all data necessary to determine compliance with required performance. These data shall provide the criteria for checking operational requirements of the detector/cooler assembly at the conclusion of the vibration test. The detector/cooler assembly shall be subjected to a simple harmonic motion having a vibratory displacement of 0.060 inch minimum peak to peak, with the frequency being changed in discrete steps of 1 Hz and maintained at least 10 seconds at each frequency over the frequency range of 10 to 55 hertz. Failure to operate shall constitute failure of this test.

4.7 Inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine compliance to requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B or Commercial, as specified (see 6.2).

5.1.1 Level A, maximum military protection.

5.1.1.1 Cleaning. Detector/cooler - Part of Viewer, Infrared AN/PAS-7 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Detector/cooler - Part of Viewer, Infrared AN/PAS-7 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

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5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Detector/cooler - Part of Viewer, Infrared AN/PAS-7. Each unit shall be individually packaged method IA-8 as follows: Cushion each unit with material conforming to PPP-C-843, type II, 1/4 inch in thickness. Place the cushioned item within a close-fitting barrier bag fabricated of material conforming to MIL-B-131, class 2. Heat seal the closure, leaving a small opening for exhausting air. Exhaust the entrapped air and complete the seal. Place the sealed bag with contents within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B, minimum military protection.

5.1.2.1 Cleaning, drying and preservation application. Detector/cooler - Part of Viewer, Infrared AN/PAS-7 shall be cleaned, dried and preserved as specified in 5.1.1.

5.1.2.2 Unit packaging. Each detector/cooler - Part of Viewer, Infrared AN/PAS-7 shall be individually packaged method IC-1 as follows: Cushion each unit with material conforming to PPP-C-843, type II, 1/4 inch in thickness. Place the cushioned item within a barrier bag fabricated of material conforming to MIL-B-121, type II. Heat seal the closure, leaving a small opening for exhausting air. Exhaust the entrapped air and complete the seal. Place the sealed bag with contents within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.3 Commercial packaging. Commercial packaging shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A, maximum military protection. A quantity of detector/cooler - Part of Viewer, Infrared AN/PAS-7, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. A quantity of the

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containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.1.1 Palletization. Palletization shall be required when; containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall be softwood conforming to NN-P-71, type IV, size 2. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film.

5.2.2 Level B, minimum military protection. A quantity of detector/cooler - Part of Viewer, Infrared AN/PAS-7, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. A quantity of the containers shall be packed with a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640, shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament type banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.2.1 Palletization. Palletization shall be required when; containers specified in 5.2.2 do not require pallet mounting; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.2, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c.

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The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film.

5.2.3 Commercial packing. Commercial packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commerical marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with FED-STD-356.

6. NOTES

6.1 Intended use. The detector/cooler, when cooled to its operating temperature and supplied with appropriate electrical biasing and optical focusing shall convert thermal information in the 3.0 to 5.0 micron spectral band to electrical signals. The detector/cooler is designed specifically for use as part of the Viewer, Infrared AN/PAS-7.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Levels of preservation-packaging and packing required (see 5.1 and 5.2).
c. When first article tests are to be conducted and the number of detector/cooler assemblies (see 3.2, 4.3 and Table II). Time frame required for submission of first article test reports and time frame required for approval of first article test reports by the contracting officer.
d. Actions required related to Group C failures.
e. Warranty clause required to insure vacuum integrity (3.5.7) and storage life (3.5.12) requirements are met for one year.

6.3 Definitions.

6.3.1 Peak responsivity. Calculate the responsivity of the chosen detector element from the data recorded in 4.6.2 and the size of the sensitive area of each element (see USAECOM Drawing SM-D-774663, Detector/Cooler), as follows:

$$\text{Peak Responsivity} = \frac{S\ C}{G\ J_{rms}\ .A_c} = \frac{\text{volts}}{\text{watt}}$$

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Where:

S = signal voltage in volts at the tester amplifier output

G = gain of tester amplifier of detector element

J_{rms} = effective rms flux density incident on the detector element in watts/cm^2

A_c = active area of detector element in cm^2

C = blackbody to peak conversion factor

$$= \frac{\int_0^\infty J\ (\lambda, T_{BB})\ d\ \lambda}{\int_c^\infty \beta\ (\lambda)\ J\ (\lambda, T_{BB})\ d\ \lambda}$$

Where:

$\beta(\lambda)$ = relative spectral response referenced to the peak responsivity of the detector-filter

$J(\lambda, T_{BB})$ = radiation output of the blackbody at temperature T_{BB}

6.3.2 Time constant. Calculate the time constant of the chosen detector elements as follows:

$$\text{Time constant} = \frac{1}{2\ \pi\ f}$$

Where:

f = chopping frequency at which the signal amplitude drops to 0.707 times its value over the low frequency region.

6.3.3 Spot size. The spot shall be defined in the following manner: The spot shall be positioned on the center of an element; then it is moved toward one edge of the element until the signal is 90 percent of its value in the center of the element. Record spot position. Continue to move the spot away from center of the element until the signal reaches 10 percent of its center (maximum) value. Record spot position. The distance between the two recorded positions is the nominal spot size.

6.3.4 Room temperature. Room temperature is defined as +23°C, ±3°C.

Custodian:
Army - EL

User activity:
Army - MU

Preparing activity:
Army - EL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R275

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-D-49088(EL) DETECTOR/COOLER ASSEMBLY FOR AN/PAS-7

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-PM 1068-68

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